

PYTHON EXPERTS (PyE)

(30th September 2023)

Members:

1. Kelvin Macharia – Currently studying API development, and backend development using Django
2. Caroline Ayieko – has received training in Data Analysis
3. Rogers Mugambi – has received training, and has some experience, in Data Science

Background

The members met on the 29th and 30th of October at 10PM EAT to deliberate on forming a learning team, and the goal and objectives of that team. All members have prior exposure to programming using Python, with various levels of proficiency.

Goal of PyE

Ensure all members are proficient in python programming.

SMART

The goal met all the requirements of the SMART protocol (Specific, Measurable, Achievable, Realistic, Timely). Beginner-level Python topics were identified, detailed and distributed amongst the members. The topics would be covered within a period of 3 months (6 meetings total, held on a fortnightly basis). The team will be meeting on the **second** and **fourth Saturday** of every month from 10 – 11PM EAT.

The topics to be covered in Phase 1 are:

- a) Data types
- b) Control flow
- c) Python installation and setting up
- d) Error handling and unit testing
- e) Functions and classes

The KPIs for this activity were identified to be:

- a) Number of times met during the 3 months period
- b) Feedback of the team members regarding level of understanding of the topics.
This will be administered through a questionnaire.

- c) Ability of the team members to complete a chosen project that will be done at the end of the 3 months

The Goal will be achieved through group self-tutoring. Each member was assigned topics within Python to research on and tutor the other members. The topics were assigned on the basis of a combination of need and expertise. Each member was assigned topics that they were insufficient in, with the expectation that this would help them actively interact with literature regarding the topics, and therefore gain understanding in them.

Work Plan

Champion	Topic	Will Include	Timelines
Caroline Ayieko	Data Types	- String manipulation and formatting - Ternary operators	6 sessions @ 1hr each By 23 rd December 2023
	Control Flow	- Enumerators - List comprehensions	
Kelvin Macharia	Python Installation and setting up	- Virtual environments - Docker	6 sessions @ 1hr each By 23 rd December 2023
	Control Flow	- Generators and iterators	
	Error handling and unit testing	- Py-tests	
Rogers Mugambi	Functions	- Creating and decorating - Using kwargs and args - Packing and unpacking - Documentation	6 sessions @ 1hr each By 23 rd December 2023
	Classes	- Why use OOP? - Creating python classes - Decorating classes - Dunder methods - Types of class methods - Small script	

Group Resources

GitHub Repo –

Project tracker –